补充材料：重叠窗口拓扑扫描的有限样本校准边界

正文引用本补充材料时按 S1–S5 编号。所有数值与正文及冻结的 canonical 汇总文件一致。

S1 完整 112 行零假设结果

下表给出 28 个唯一零假设配置 × 2 表示 × 2 方法共 112 个逐格评价单元的拒绝率（每格 $n = 2500$）、95% Wilson 区间（无连续性修正）与四类校准状态。列含义：DGP 为数据生成过程；ρ 为原序列依赖；s 为步长；T、M 为观测长度与窗口数；分类取实用校准 / 实质保守 / 实质反保守 / 边界或未定。本表的分类列即正文第 5.2 节所称的完整状态转换清单。

DGP	ρ	s	T	M	表示	方法	拒绝率	95% Wilson	分类
N-AR	0.8	64	12288	192	BPI	M0	0.0524	[0.0443,0.0618]	实用校准
N-AR	0.8	64	12288	192	BPI	M1	0.0276	[0.0219,0.0348]	实质保守
N-AR	0.8	64	12288	192	MOMENTS	M0	0.0432	[0.0359,0.0519]	实用校准
N-AR	0.8	64	12288	192	MOMENTS	M1	0.0216	[0.0166,0.0281]	实质保守
N-AR	0	32	6176	192	BPI	M0	0.3044	[0.2867,0.3227]	实质反保守
N-AR	0	32	6176	192	BPI	M1	0.0492	[0.0414,0.0584]	实用校准
N-AR	0	32	6176	192	MOMENTS	M0	0.4376	[0.4183,0.4571]	实质反保守
N-AR	0	32	6176	192	MOMENTS	M1	0.0404	[0.0334,0.0488]	边界或未定
N-CIRCLE	0	8	1592	192	BPI	M0	0.8816	[0.8683,0.8937]	实质反保守
N-CIRCLE	0	8	1592	192	BPI	M1	0.0436	[0.0363,0.0523]	实用校准
N-CIRCLE	0	8	1592	192	MOMENTS	M0	0.9988	[0.9965,0.9996]	实质反保守
N-CIRCLE	0	8	1592	192	MOMENTS	M1	0.0700	[0.0606,0.0807]	边界或未定
N-AR	0	8	1592	192	BPI	M0	0.9712	[0.9639,0.9771]	实质反保守
N-AR	0	8	1592	192	BPI	M1	0.0808	[0.0707,0.0921]	实质反保守
N-AR	0	8	1592	192	MOMENTS	M0	0.9976	[0.9948,0.9989]	实质反保守
N-AR	0	8	1592	192	MOMENTS	M1	0.0712	[0.0618,0.0820]	边界或未定
N-CIRCLE	0.8	8	1592	192	BPI	M0	0.9384	[0.9283,0.9472]	实质反保守
N-CIRCLE	0.8	8	1592	192	BPI	M1	0.0496	[0.0418,0.0588]	实用校准
N-CIRCLE	0.8	8	1592	192	MOMENTS	M0	0.9968	[0.9937,0.9984]	实质反保守
N-CIRCLE	0.8	8	1592	192	MOMENTS	M1	0.0792	[0.0692,0.0904]	实质反保守
N-AR	0	64	3120	48	BPI	M0	0.0608	[0.0521,0.0709]	边界或未定
N-AR	0	64	3120	48	BPI	M1	0.0088	[0.0058,0.0133]	实质保守

DGP	ρ	s	T	M	表示	方法	拒绝率	95% Wilson	分类
N-AR	0	64	3120	48	MOMENTS	M0	0.0584	[0.0499,0.0683]	边界或未定
N-AR	0	64	3120	48	MOMENTS	M1	0.0020	[0.0009,0.0047]	实质保守
N-CIRCLE	0	64	12288	192	BPI	M0	0.0528	[0.0447,0.0623]	实用校准
N-CIRCLE	0	64	12288	192	BPI	M1	0.0244	[0.0190,0.0312]	实质保守
N-CIRCLE	0	64	12288	192	MOMENTS	M0	0.0532	[0.0451,0.0627]	实用校准
N-CIRCLE	0	64	12288	192	MOMENTS	M1	0.0228	[0.0176,0.0294]	实质保守
N-AR	0.8	64	3120	48	BPI	M0	0.0548	[0.0465,0.0644]	实用校准
N-AR	0.8	64	3120	48	BPI	M1	0.0108	[0.0074,0.0157]	实质保守
N-AR	0.8	64	3120	48	MOMENTS	M0	0.0492	[0.0414,0.0584]	实用校准
N-AR	0.8	64	3120	48	MOMENTS	M1	0.0056	[0.0033,0.0094]	实质保守
N-CIRCLE	0.8	64	3120	48	BPI	M0	0.0552	[0.0469,0.0649]	实用校准
N-CIRCLE	0.8	64	3120	48	BPI	M1	0.0056	[0.0033,0.0094]	实质保守
N-CIRCLE	0.8	64	3120	48	MOMENTS	M0	0.0616	[0.0528,0.0717]	边界或未定
N-CIRCLE	0.8	64	3120	48	MOMENTS	M1	0.0032	[0.0016,0.0063]	实质保守
N-AR	0.8	8	3120	383	BPI	M0	0.9480	[0.9386,0.9560]	实质反保守
N-AR	0.8	8	3120	383	BPI	M1	0.1180	[0.1059,0.1312]	实质反保守
N-AR	0.8	8	3120	383	MOMENTS	M0	0.9912	[0.9867,0.9942]	实质反保守
N-AR	0.8	8	3120	383	MOMENTS	M1	0.1064	[0.0949,0.1191]	实质反保守
N-CIRCLE	0	64	3120	48	BPI	M0	0.0472	[0.0396,0.0562]	实用校准
N-CIRCLE	0	64	3120	48	BPI	M1	0.0060	[0.0036,0.0099]	实质保守
N-CIRCLE	0	64	3120	48	MOMENTS	M0	0.0520	[0.0440,0.0614]	实用校准
N-CIRCLE	0	64	3120	48	MOMENTS	M1	0.0024	[0.0011,0.0052]	实质保守
N-CIRCLE	0	16	3120	192	BPI	M0	0.4384	[0.4191,0.4579]	实质反保守
N-CIRCLE	0	16	3120	192	BPI	M1	0.0480	[0.0403,0.0571]	实用校准
N-CIRCLE	0	16	3120	192	MOMENTS	M0	0.9376	[0.9274,0.9464]	实质反保守
N-CIRCLE	0	16	3120	192	MOMENTS	M1	0.0760	[0.0663,0.0871]	实质反保守
N-AR	0.8	32	3120	96	BPI	M0	0.2668	[0.2498,0.2845]	实质反保守
N-AR	0.8	32	3120	96	BPI	M1	0.0352	[0.0287,0.0432]	边界或未定
N-AR	0.8	32	3120	96	MOMENTS	M0	0.3336	[0.3154,0.3523]	实质反保守
N-AR	0.8	32	3120	96	MOMENTS	M1	0.0228	[0.0176,0.0294]	实质保守
N-AR	0	16	3120	192	BPI	M0	0.7356	[0.7180,0.7525]	实质反保守
N-AR	0	16	3120	192	BPI	M1	0.0672	[0.0580,0.0777]	边界或未定
N-AR	0	16	3120	192	MOMENTS	M0	0.9136	[0.9019,0.9240]	实质反保守
N-AR	0	16	3120	192	MOMENTS	M1	0.0680	[0.0588,0.0785]	边界或未定
N-CIRCLE	0.8	32	6176	192	BPI	M0	0.1556	[0.1419,0.1703]	实质反保守
N-CIRCLE	0.8	32	6176	192	BPI	M1	0.0280	[0.0222,0.0352]	边界或未定
N-CIRCLE	0.8	32	6176	192	MOMENTS	M0	0.4560	[0.4366,0.4756]	实质反保守
N-CIRCLE	0.8	32	6176	192	MOMENTS	M1	0.0472	[0.0396,0.0562]	实用校准
N-CIRCLE	0	32	6176	192	BPI	M0	0.1088	[0.0972,0.1216]	实质反保守
N-CIRCLE	0	32	6176	192	BPI	M1	0.0284	[0.0226,0.0357]	边界或未定
N-CIRCLE	0	32	6176	192	MOMENTS	M0	0.4984	[0.4788,0.5180]	实质反保守
N-CIRCLE	0	32	6176	192	MOMENTS	M1	0.0332	[0.0269,0.0410]	边界或未定
N-CIRCLE	0.8	32	3120	96	BPI	M0	0.1584	[0.1446,0.1732]	实质反保守
N-CIRCLE	0.8	32	3120	96	BPI	M1	0.0184	[0.0138,0.0245]	实质保守
N-CIRCLE	0.8	32	3120	96	MOMENTS	M0	0.4144	[0.3952,0.4338]	实质反保守
N-CIRCLE	0.8	32	3120	96	MOMENTS	M1	0.0276	[0.0219,0.0348]	实质保守
N-AR	0	32	3120	96	BPI	M0	0.2776	[0.2604,0.2955]	实质反保守
N-AR	0	32	3120	96	BPI	M1	0.0288	[0.0229,0.0361]	边界或未定
N-AR	0	32	3120	96	MOMENTS	M0	0.4260	[0.4067,0.4455]	实质反保守

DGP	ρ	s	T	M	表示	方法	拒绝率	95% Wilson	分类
N-AR	0	32	3120	96	MOMENTS	M1	0.0212	[0.0162,0.0276]	实质保守
N-AR	0.8	16	3120	192	BPI	M0	0.6688	[0.6501,0.6870]	实质反保守
N-AR	0.8	16	3120	192	BPI	M1	0.0788	[0.0689,0.0900]	实质反保守
N-AR	0.8	16	3120	192	MOMENTS	M0	0.7928	[0.7765,0.8082]	实质反保守
N-AR	0.8	16	3120	192	MOMENTS	M1	0.0628	[0.0539,0.0730]	边界或未定
N-CIRCLE	0	8	3120	383	BPI	M0	0.8900	[0.8771,0.9017]	实质反保守
N-CIRCLE	0	8	3120	383	BPI	M1	0.0808	[0.0707,0.0921]	实质反保守
N-CIRCLE	0	8	3120	383	MOMENTS	M0	0.9996	[0.9977,0.9999]	实质反保守
N-CIRCLE	0	8	3120	383	MOMENTS	M1	0.1320	[0.1193,0.1458]	实质反保守
N-AR	0.8	8	1592	192	BPI	M0	0.9324	[0.9219,0.9416]	实质反保守
N-AR	0.8	8	1592	192	BPI	M1	0.0892	[0.0786,0.1010]	实质反保守
N-AR	0.8	8	1592	192	MOMENTS	M0	0.9816	[0.9755,0.9862]	实质反保守
N-AR	0.8	8	1592	192	MOMENTS	M1	0.0676	[0.0584,0.0781]	边界或未定
N-CIRCLE	0.8	16	3120	192	BPI	M0	0.5416	[0.5220,0.5611]	实质反保守
N-CIRCLE	0.8	16	3120	192	BPI	M1	0.0524	[0.0443,0.0618]	实用校准
N-CIRCLE	0.8	16	3120	192	MOMENTS	M0	0.8796	[0.8663,0.8918]	实质反保守
N-CIRCLE	0.8	16	3120	192	MOMENTS	M1	0.0848	[0.0745,0.0964]	实质反保守
N-CIRCLE	0.8	8	3120	383	BPI	M0	0.9432	[0.9334,0.9516]	实质反保守
N-CIRCLE	0.8	8	3120	383	BPI	M1	0.0824	[0.0723,0.0938]	实质反保守
N-CIRCLE	0.8	8	3120	383	MOMENTS	M0	0.9964	[0.9932,0.9981]	实质反保守
N-CIRCLE	0.8	8	3120	383	MOMENTS	M1	0.1412	[0.1281,0.1554]	实质反保守
N-CIRCLE	0.8	64	12288	192	BPI	M0	0.0528	[0.0447,0.0623]	实用校准
N-CIRCLE	0.8	64	12288	192	BPI	M1	0.0244	[0.0190,0.0312]	实质保守
N-CIRCLE	0.8	64	12288	192	MOMENTS	M0	0.0516	[0.0436,0.0610]	实用校准
N-CIRCLE	0.8	64	12288	192	MOMENTS	M1	0.0236	[0.0183,0.0303]	实质保守
N-AR	0	64	12288	192	BPI	M0	0.0504	[0.0425,0.0597]	实用校准
N-AR	0	64	12288	192	BPI	M1	0.0224	[0.0173,0.0290]	实质保守
N-AR	0	64	12288	192	MOMENTS	M0	0.0492	[0.0414,0.0584]	实用校准
N-AR	0	64	12288	192	MOMENTS	M1	0.0192	[0.0145,0.0254]	实质保守
N-AR	0	8	3120	383	BPI	M0	0.9756	[0.9688,0.9810]	实质反保守
N-AR	0	8	3120	383	BPI	M1	0.1136	[0.1017,0.1266]	实质反保守
N-AR	0	8	3120	383	MOMENTS	M0	0.9984	[0.9959,0.9994]	实质反保守
N-AR	0	8	3120	383	MOMENTS	M1	0.1184	[0.1063,0.1317]	实质反保守
N-AR	0.8	32	6176	192	BPI	M0	0.2640	[0.2471,0.2816]	实质反保守
N-AR	0.8	32	6176	192	BPI	M1	0.0408	[0.0337,0.0493]	边界或未定
N-AR	0.8	32	6176	192	MOMENTS	M0	0.3440	[0.3256,0.3628]	实质反保守
N-AR	0.8	32	6176	192	MOMENTS	M1	0.0312	[0.0251,0.0388]	边界或未定
N-CIRCLE	0	32	3120	96	BPI	M0	0.1200	[0.1078,0.1333]	实质反保守
N-CIRCLE	0	32	3120	96	BPI	M1	0.0188	[0.0142,0.0249]	实质保守
N-CIRCLE	0	32	3120	96	MOMENTS	M0	0.4820	[0.4625,0.5016]	实质反保守
N-CIRCLE	0	32	3120	96	MOMENTS	M1	0.0164	[0.0121,0.0222]	实质保守

S2 完整 16 行备择结果

下表给出两个场景 × 4 条件 × 2 表示共 16 个“条件–表示”单元的尺寸校正功效、归一化定位误差、联合成功率与名义功效（每格 $n = 1600$，临界秩 2376）。列含义：ρ 为原序列依赖；s 为步长；J_0、J_1 分别为 M0、M1 的联合成功率；β_0、β_1 分别为 M0、M1 的名

义功效。

场景	ρ	s	表示	尺寸校正功效	nMAE	J_0	J_1	β_0	β_1
形状变化	0	64	BPI	1.00000	0.00001	0.9988	0.9988	1.0000	1.0000
形状变化	0	64	MOMENTS	0.07125	0.21158	0.0006	0.0006	0.0531	0.0206
均值平移	0	64	BPI	0.05312	0.20918	0.0000	0.0000	0.0537	0.0225
均值平移	0	64	MOMENTS	1.00000	0.00003	0.9944	0.9944	1.0000	1.0000
形状变化	0.8	16	BPI	1.00000	0.00772	0.9912	0.9912	1.0000	1.0000
形状变化	0.8	16	MOMENTS	0.04562	0.20186	0.0350	0.0019	0.9294	0.0825
均值平移	0.8	64	BPI	0.04688	0.21402	0.0006	0.0006	0.0419	0.0187
均值平移	0.8	64	MOMENTS	1.00000	0.00132	0.8119	0.8119	1.0000	1.0000
均值平移	0	16	BPI	0.05437	0.18659	0.0219	0.0037	0.4437	0.0406
均值平移	0	16	MOMENTS	1.00000	0.00284	0.9988	0.9988	1.0000	1.0000
均值平移	0.8	16	BPI	0.06062	0.20040	0.0125	0.0000	0.5819	0.0512
均值平移	0.8	16	MOMENTS	1.00000	0.00902	0.8156	0.8156	1.0000	1.0000
形状变化	0.8	64	BPI	1.00000	0.00035	0.9456	0.9456	1.0000	1.0000
形状变化	0.8	64	MOMENTS	0.06688	0.18595	0.0000	0.0000	0.0981	0.0413
形状变化	0	16	BPI	1.00000	0.00828	1.0000	1.0000	1.0000	1.0000
形状变化	0	16	MOMENTS	0.06438	0.20720	0.0219	0.0013	0.9500	0.0731

S3 主对比与表示差自助法的完整规格

主对比 P1–P3：使用 10,000 次按数据生成过程分层的整簇自助法；同一主重复编号下的全部相关条件、表示和方法随簇共同重采样。单侧自助 p 值按加一规则 $p = (1 + \#\{\text{尾部}\})/10001$ 计算，三个单侧 p 值经 Holm step-down 校正。边际 95% 区间为最近秩（nearest-rank）分位数定义的 basic bootstrap interval。

表示差 Δ：通过两层整簇自助法估计，每层 2000 次。重采样单位为主复制：零假设层按 N-CIRCLE 主复制整簇重采样以重算经验临界值，备择层按备择主复制整簇重采样以重算功效；两层分别独立重采样。BPI 与 MOMENTS 使用同一重采样索引成对移动；四个 (ρ, s) 条件在每个场景内先计算各自的 BPI 减 MOMENTS 功效差，再等权平均得到场景差。区间采用最近秩定义的 2000 次百分位区间（2.5% 与 97.5% 分位数）。

S4 完整 16×4 热图

图 1 给出 16 个条件（DGP×$\rho \times s$）× 4 列（BPI/M0、BPI/M1、MOMENTS/M0、MOMENTS/M1）的完整逐格 I 类错误率，两个面板分别对应 fixed-M 与 fixed-T。色标 $[0, 1]$，格内标注四位小数。

S5 代码与数据可用性

复现本文结果的代码、配置、种子清单、完整逐格数据与全部生成脚本由仓库提供：

- 代码仓库：`https://github.com/dw0911/owtda-calibration`

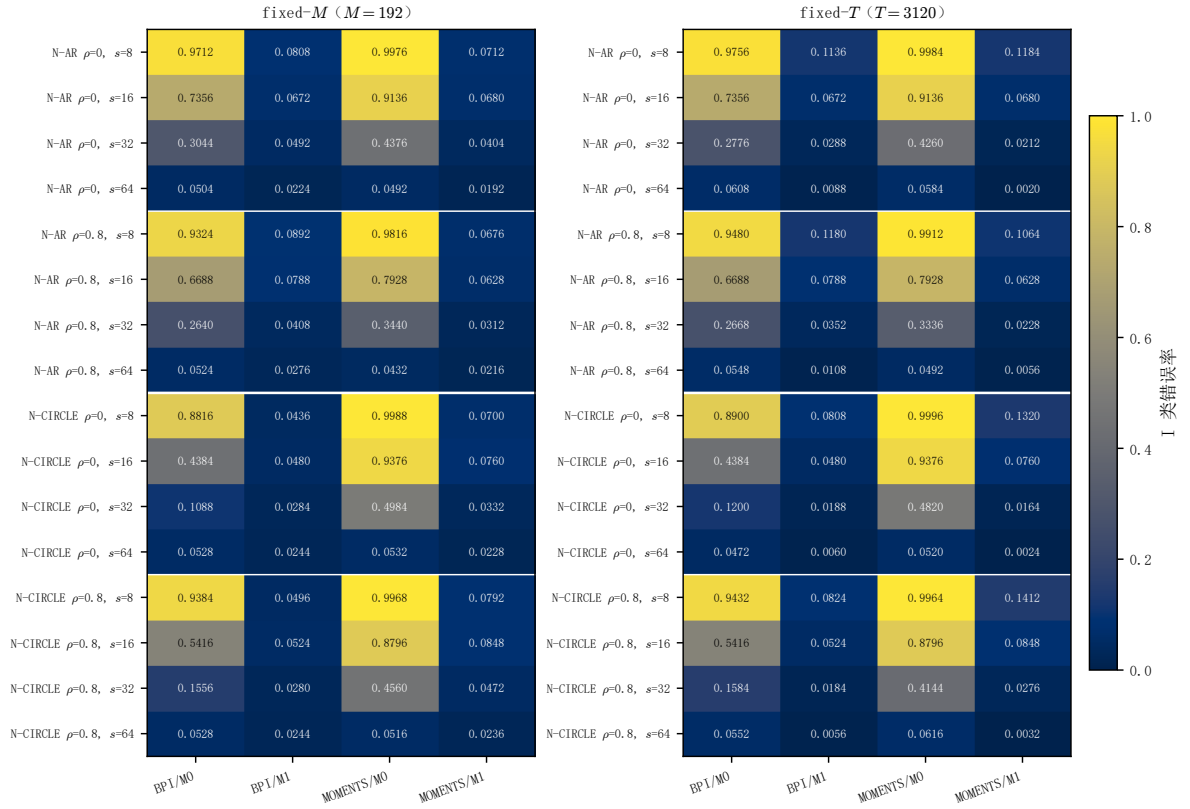


图 1: 完整 16×4 逐格 I 类错误率热图。两面板分别为 fixed-M（$M = 192$）与 fixed-T（$T = 3120$）；16 行按 DGP×$\rho \times s$ 排列，4 列为 BPI/M0、BPI/M1、MOMENTS/M0、MOMENTS/M1。

- 本稿对应 git 标签 `v1.1.0-paper`
- 全部合成数据由该版本代码与固定随机种子生成；独立数据 DOI 尚在建立中，未作为已提供材料列出